LESSON 5.9
SOLVING PROBLEMS INVOLVING DIMENSIONS AND AREAS OF GEOMETRIC FIGURES

LESSON INTRODUCTION

MA.7.GR.1.5 Solve mathematical and real-world problems involving dimensions and areas of geometric figures, including scale drawings and scale factors.

LEARNING TARGETS

- I can solve mathematical and real-world problems involving dimensions and areas of geometric figures.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.8.GR.2.4 Solve mathematical and real-world problems involving proportional relationships between similar triangles.

ESSENTIAL QUESTIONS

- How does scale factor relate to the area of scaled figures?
- How can I determine the area of a figure given only a scale factor, an original area, or set of dimensions?

LESSON NARRATIVE

This lesson extends students' understanding of scale factor as it applies to dimensions and area when solving real-world problems. As students build a deeper understanding of how to apply the concepts, they have been learning to solve problems, the scales used in this lesson include fractions. Students continue to use equivalent-ratio thinking in solving the problems presented in this lesson collaboratively before practicing on their own.

COMPONENT SUMMARY

5.9.1 WARM-UP *(~5 MIN.)*

Create a reasonable estimate to scale down the Great Pyramid

5.9.2 GUIDED PRACTICE *(15-20 MIN.)*

Extension of scale factor and its relationship with the area of geometric figures

5.9.3 YOUR TURN! *(10-15 MIN.)*

Use scale factor to solve problems involving area or dimensions of scaled figures

5.9.4 WRAP-UP *(~5 MIN.)*

Determine area based on given various scale factors

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- N/A

MATERIALS

- N/A

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may not understand why the scale factor can be squared when applying to area.
- Students may not recognize applying a scale factor of 1 would result in a model the same size as the original.

THINGS TO CONSIDER

- Reinforce strategies for students to organize their thinking as they solve problems.
- Evaluating square roots is addressed in a Grade 8 benchmark. To complete question 2 in the 5.9.3 Your Turn!, students do not need to know how to find the square root of 81. Rather, they can think about what scale factor multiplied by itself would result in 81.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

As students work on different problems in 5.9.3 Your Turn!, consider allowing students to discuss different ways of solving problems, specifically question 3, where students could either determine area of the scale first or determine the dimensions of the real figure first.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.6.1 Reasonableness

An ongoing theme in lessons of this unit is the idea of determining if scale drawings or calculations are reasonable. Students examine in this way beginning with the 5.9.1 Warm-Up when they look to estimate a reasonable scale factor to use if they wanted to create a smaller scale of the Great Pyramid.

DIFFERENTIATE INSTRUCTION

The On-Ramp to 7th Grade Accelerated Math video and practice tools offer instruction and practice for a variety of previously learned skills to support all students. Consider having students preview this lesson by watching the On-Ramp video or having the video available for students to reference throughout this lesson.

- Perimeter, Area, and Volume: Which Should You Use?
- Using Area Formula for Rectangles
- Finding Area of Rectilinear Shapes

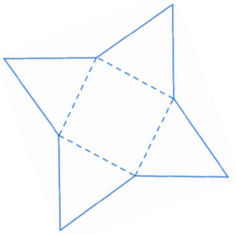
5.9.1 WARM-UP: BETWEEN TWO NUMBERS

TEACHER MOVES

Consider giving students the opportunity to discuss what some unreasonable scales might be. Sometimes by determining what won't work, students can narrow their choices for what might work.

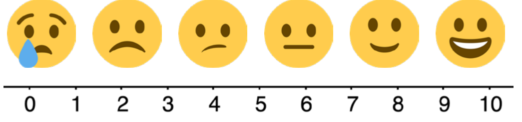
5.9.1 Warm-Up: Between Two Numbers

The Great Pyramid of Giza, the oldest of the Seven Wonders of the Ancient World, took about 27 years to build and was completed around 2560 B.C. It is 146.7 meters (m) tall and has a volume of $2{,}583{,}283$ cubic meters (m^3).



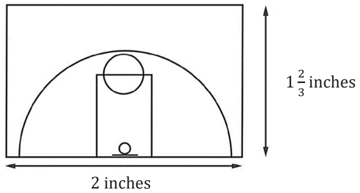
1. To build a scale model of the Great Pyramid that is 3000 times smaller than the original using regular 8.5 by 11 inches (in.) paper, what is a reasonable scale that could be used?
   Answers will vary. I would use a scale of $\frac{1 \text{ in}}{100 \text{ m}}$

2. Circle the number that best describes your confidence in your estimate.
   Answers will vary.



5.9.2 Guided Practice: Scale and Area

1. Vincent's scale drawing of the basketball half-court he proposed to City Council is shown. The width is $1\frac{2}{3}$ in. and the length is 2 in.



After reviewing Vincent's work, the council agreed that using a scale of 15 feet (ft) to 1 in. would be acceptable. The council will use the area of the court to calculate the cost of materials in square feet.

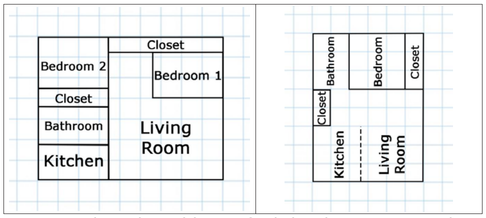
Use the scale $\frac{15\text{ ft}}{1\text{ in.}}$ and the drawing to determine the area of the actual basketball court.

$$\left(2\ in. \times 1\frac{2}{3}\ in.\right) \times \left(\frac{15\text{ ft}}{1\text{ in.}}\right)^2 = 750\ ft^2$$

Using the scale and model provided, we can conclude that the area of the actual basketball court is 750 ft².

2. The models of two apartments are shown with the scales used to create them. Each gridline represents 1 centimeter (cm).

Suburban Apartment	**City Apartment**
Scale: $\frac{5.5\text{ feet}}{1\text{ centimeter}}$	**Scale:** $\frac{4\text{ feet}}{1\text{ centimeter}}$



a. Complete the tables to find the dimensions and areas of both apartments.

Suburban Apartment			
Scale	**Length**	**Width**	**Area**
1	9 cm	7 cm	63cm²
$\frac{5.5\text{ ft}}{1\text{ cm}}$	49.5 ft	38.5 ft	1,905.75 ft²

City Apartment			
Scale	**Length**	**Width**	**Area**
1	6 cm	8 cm	48 cm²
$\frac{4\text{ ft}}{1\text{ cm}}$	24 ft	32 ft	768 ft²

b. Which apartment is larger?
The suburban apartment is larger than the city apartment.

c. Immanuel is looking for an apartment with as much closet space as possible. Explain which apartment you would recommend for him.
The suburban apartment has a total of
$$\left(3.5\ cm * 0.75\ cm * \frac{5.5\ ft}{1\ cm} * \frac{5.5\ ft}{1\ cm}\right) + \left(5.6\ cm * 0.75\ cm * \frac{5.5\ ft}{1\ cm} * \frac{5.5\ ft}{1\ cm}\right) = 206.46\ ft^2$$
of closet space (student work will vary).
The city apartment has
$$\left(1\ cm * 2\ cm * \frac{4\ ft}{1\ cm} * \frac{4\ ft}{1\ cm}\right) + \left(1cm * 3cm * \frac{4\ ft}{1\ cm} * \frac{4\ ft}{1\ cm}\right) = 80\ ft^2$$
of closet space.
I would recommend the suburban apartment for Immanuel since it has the most closet space.

5.9.2 GUIDED PRACTICE: SCALE AND AREA

DIFFERENTIATE INSTRUCTION

Students should be familiar with this context from the previous lesson in which they found the perimeter of the basketball court. This question instead focuses on finding area using scale factor.

TEACHER MOVES

Some students forget to square the scale factor before using it to find the area. They could also solve this problem by using the scale factor to determine each dimension on the actual court and then multiplying to find the area.

If students are struggling to get started, consider further scaffolding this section through questioning: How can you find the length of the actual basketball court using the scale factor? The width? How can you use the length and width of the actual basketball court to find its area? Why is the scale factor squared when working with area?

DIFFERENTIATE INSTRUCTION

Consider splitting students into groups or splitting the class in half, with some groups exploring the suburban apartment and the other groups exploring the city apartment. Students could be tasked with calculating the necessary dimensions and then comparing and checking work with a partner before working together on questions 2b and 2c.

5.9.3 YOUR TURN!

ENRICHMENT

Students may struggle to explain whether they agree or disagree with Shea. Ask questions to further scaffold, if needed:

- What is the relationship between scale factor and area?
- How can you find the area of the actual restaurant given the area of the scale drawing and the scale factor?

TEACHER MOVES

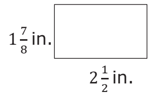
Consider discussing possible different strategies such as using the scale factor to determine the dimensions and then the area or to determine the area of the scale and then using the scale factor with the area to determine actual area.

5.9.3 Your Turn!

1. On a map with a scale of 1 inch (in.) to 12 feet (ft), the area of a restaurant is 60 square inches (in^2). Shea says that the actual area of the restaurant is 720 square feet (ft^2). Explain why you agree or disagree with Shea.
   I disagree with Shea. She should have squared the scale factor when finding the area of the actual restaurant.
   $$60 \times (12)^2 = 8640 \textit{ square feet}$$

2. Hunter finds that the area of a house on a scale drawing is 25 in^2. The actual area of the house is $2{,}025$ ft^2. What is the scale of the drawing?
   $$\frac{2{,}025}{25} = 81$$
   What scale factor multiplied by itself equals 81?
   $$9 \times 9 = 81$$
   The scale used for the drawing was $1 \textit{ inch} : 9 \textit{ feet}$.

3. A scale drawing of the floor plan of Mariko's workshop is shown. If the scale is $\frac{40 \text{ ft}}{1 \text{ in.}}$, find the area, in square feet, of Mariko's actual workshop.



   $$\left(1\frac{7}{8} in. \times 2\frac{1}{2} in.\right) \times \left(\frac{40\ ft}{1\ in.}\right)^2$$
   $$= 7{,}500\ ft^2$$

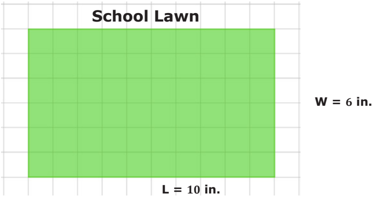
   or

   $$\left(1\frac{7}{8}\text{in.} \times \frac{40 \text{ ft}}{1 \text{ in.}}\right) \times \left(2\frac{1}{2}\text{in.} \times \frac{40 \text{ ft}}{1 \text{ in.}}\right) = 7{,}500\ ft^2$$

   Mariko's workshop has an area of $7{,}500\ ft^2$.

4. The greenhouse club needs to determine how much seed to buy to cover the lawn in the school courtyard. Anthony suggests they use his school map to calculate the area that will need to be covered in seed.

   Each grid on the map represents 1 in. Anthony measures the rectangular area on the map and finds the length (L) to be 10 in. and the width (W) to be 6 in.



   The scale of the map is $\frac{7 \text{ ft}}{1 \text{ in.}}$.

   What is the actual area in square feet?
   $$(6\ in. \times 10\ in.) \times \left(\frac{7 \text{ ft}}{1 \text{ in.}}\right)^2 = 2{,}940\ ft^2$$

   The area of the school courtyard is $2{,}940\ ft^2$.

5.9.4 Wrap-Up: Cool Down

1. What is the relationship between a model's scale factor and area?
   The model's area is multiplied by the square of the scale factor to obtain the area of the actual figure.

2. Use that relationship to complete the following table.

Scale Factor	Area (unit^2)
1	56
$\frac{1}{5}$	2.24
$\frac{1}{4}$	3.5
4	896

5.9.4 WRAP-UP: COOL DOWN

ENRICHMENT

Students may struggle to recognize where to find the area of the original figure; the original figure has a scale factor of 1.